Aliasing and the Sampling Theorem Simplified

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Key Concepts

1) Three key facts for understanding sampling and aliasing:

    a) Arbitrary signals can be expressed as a sum of sinusoids using the Fourier transform.

    b) A continuous-time sinusoid with frequency Ω maps to a discrete-time sinusoid of frequency $\omega = \Omega T$ where T is the sampling interval.

    c) Discrete-time sinusoids are only unique over a 2π interval of ω. We will use $-\pi < \omega \leq \pi$.

2) The range $-\pi < \omega \leq \pi$ corresponds to $-\frac{\pi}{T} < \Omega \leq \frac{\pi}{T}$. Aliasing results because frequencies $\Omega > \frac{\pi}{T}$ or $\Omega \leq -\frac{\pi}{T}$ map into the same discrete-time frequency range $-\pi < \omega \leq \pi$.

3) We cannot uniquely determine the continuous-time frequency Ω given the discrete-time frequency ω unless we have prior knowledge about the range of the continuous-time frequency, such as $|\Omega| < \frac{\pi}{T}$.

4) The sampling theorem states that if $x(t)$ is band limited with maximum frequency W rads/sec, then $x(t)$ is uniquely described by its samples $x(nT)$ provided $W < \frac{\pi}{T}$.

5) The Fourier transform $X_s(\Omega)$ of a sampled signal $x[n] = x(nT)$ is obtained by defining $X_s(\Omega) = X(e^{j\omega})\Big|_{\omega=\Omega T}$ where $X(e^{j\omega})$ is the discrete-time Fourier transform of $x[n]$. If $X(\Omega)$ is the Fourier transform of $x(t)$ and $\Omega_s = \frac{2\pi}{T}$, then

$$X_s(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(\Omega - k\Omega_s)$$

Video Screenshots

Three Key Facts-

1) An arbitrary signal can be expressed as a "sum" of sinusoids (Fourier Transform)

2) Sampling at intervals of T sec relates CT frequency Ω to DT frequency ω as

$$\omega \text{ (rads)} = \Omega \left(\frac{\text{rads}}{\text{sec}}\right) \cdot T \text{ (sec)}$$

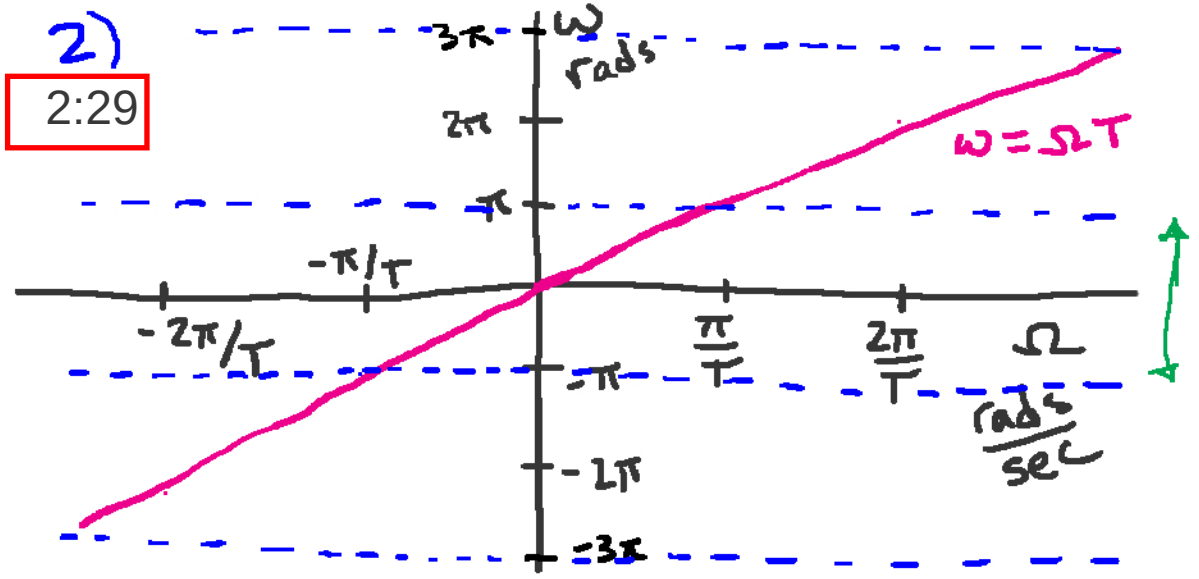
$e^{j\Omega t}$ → [Sample $t = nT$] → $e^{j\Omega Tn}$ → $e^{j\omega n}$ for $\omega = \Omega T$

3) DT sinusoids are only unique over a 2π interval of ω

$$e^{j(\omega + l2\pi)n} = e^{j\omega n} e^{j2\pi ln} = e^{j\omega n}$$

2)

2:29

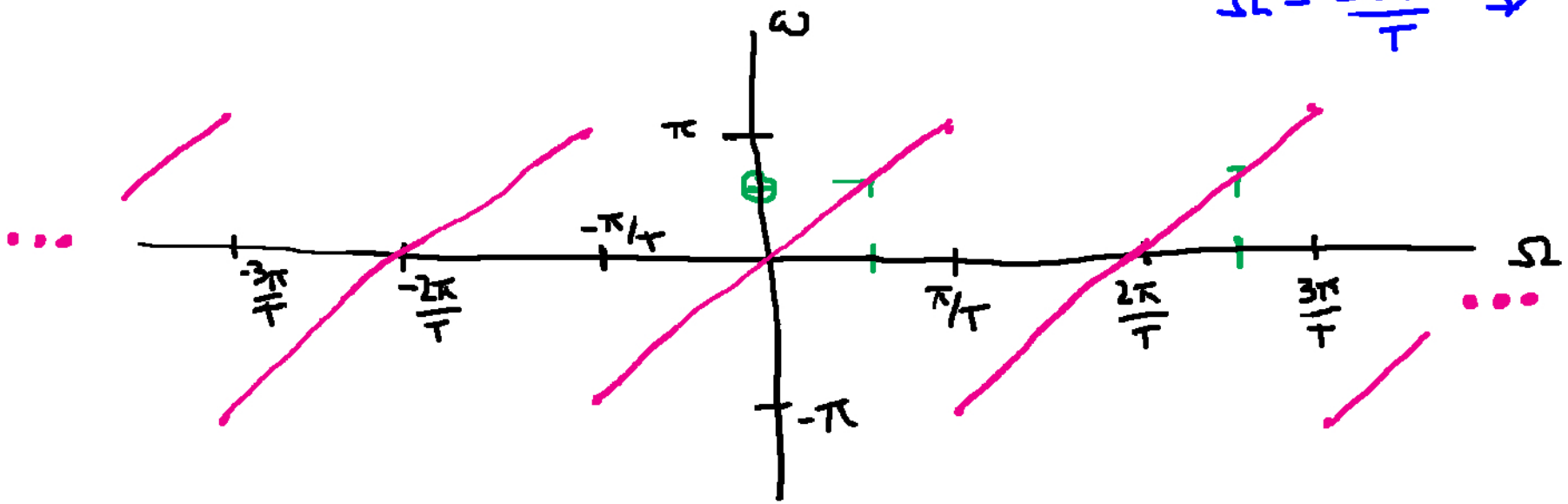


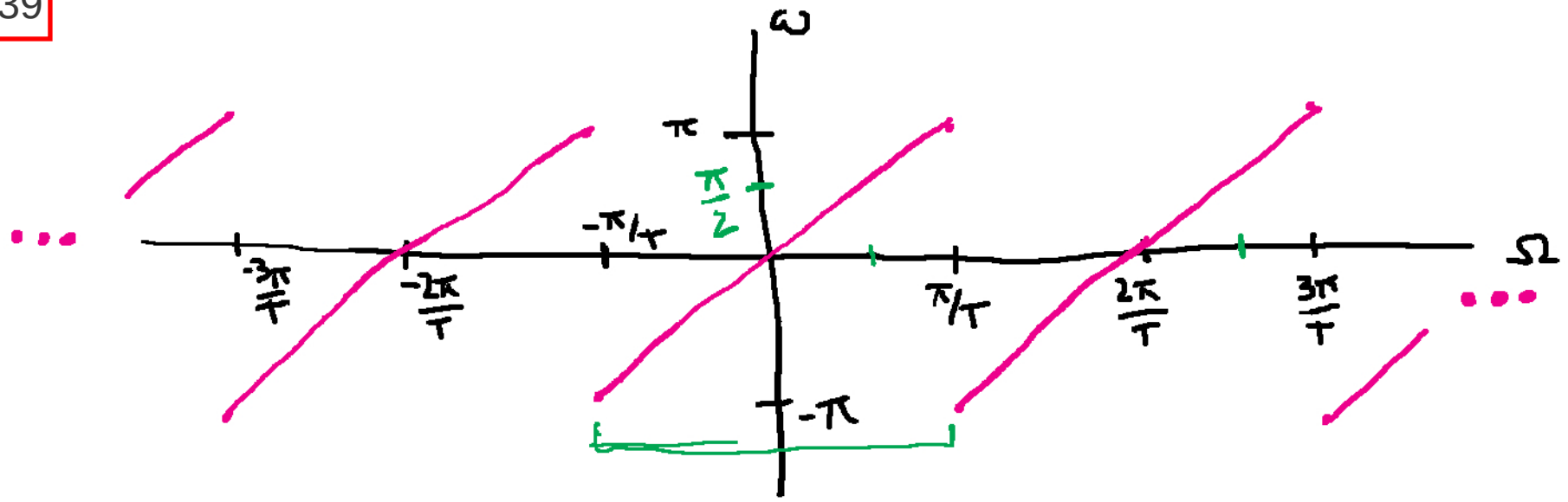
3) $e^{j\omega n}$ only distinct on a 2π interval of ω

$\Rightarrow$ Use $-\pi < \omega < \pi$

$\Omega = \frac{0.5\pi}{T} \Rightarrow \omega = \frac{\pi}{2}$

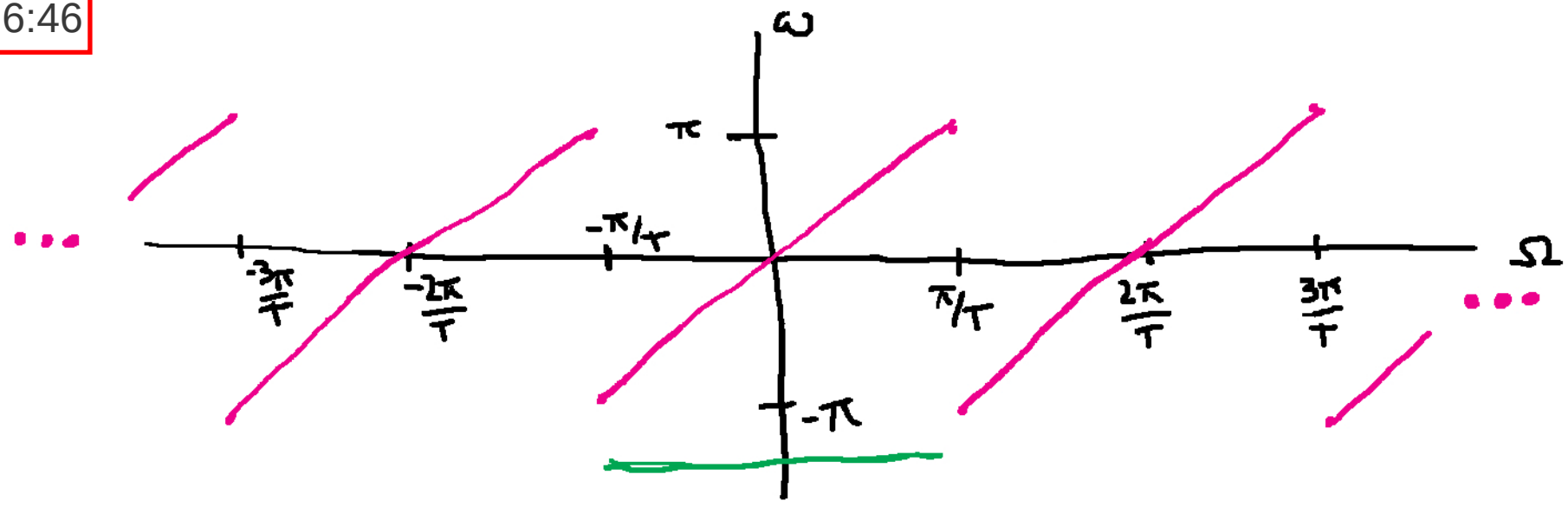
$\Omega = \frac{2.5\pi}{T} \Rightarrow \omega = \pi/2$





Aliasing – CT sinusoids ($|\Omega| > \frac{\pi}{T}$) appear at "incorrect" DT frequencies ω

Reconstruction – given DT sinusoid (ω), cannot uniquely determine the CT sinusoid (Ω) unless we know $|\Omega| < \pi/T$



Sampling Theorem (Nyquist): If $x(t) \overset{FT}{\longleftrightarrow} X(\Omega)$ is bandlimited $\Rightarrow$ $|X(\Omega)| = 0$, $|\Omega| > W$ rads/sec,

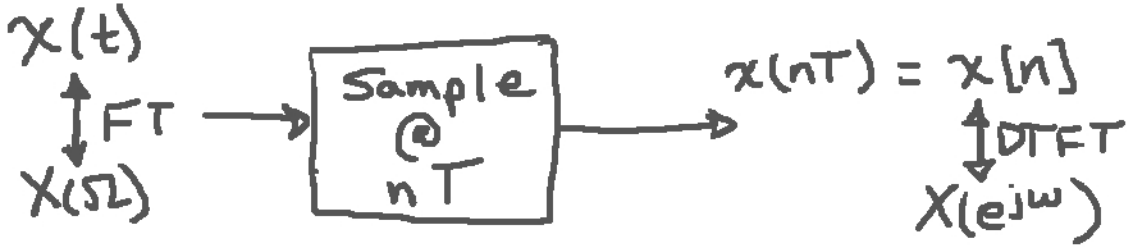
Then $x(t)$ is uniquely described by its samples $x(nT)$ provided $W < \pi/T$. $T < \frac{\pi}{W}$

$\frac{2\pi}{T}$: sampling frequency ($\frac{1}{T}$ Hz)

W: Nyquist frequency

2W: Nyquist rate

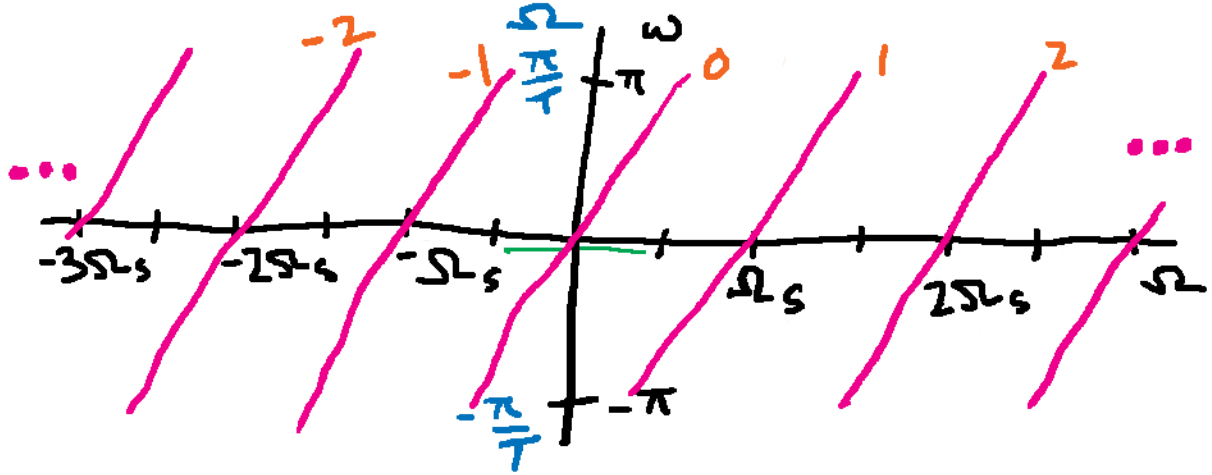
Fourier Transform of Sampled Signal

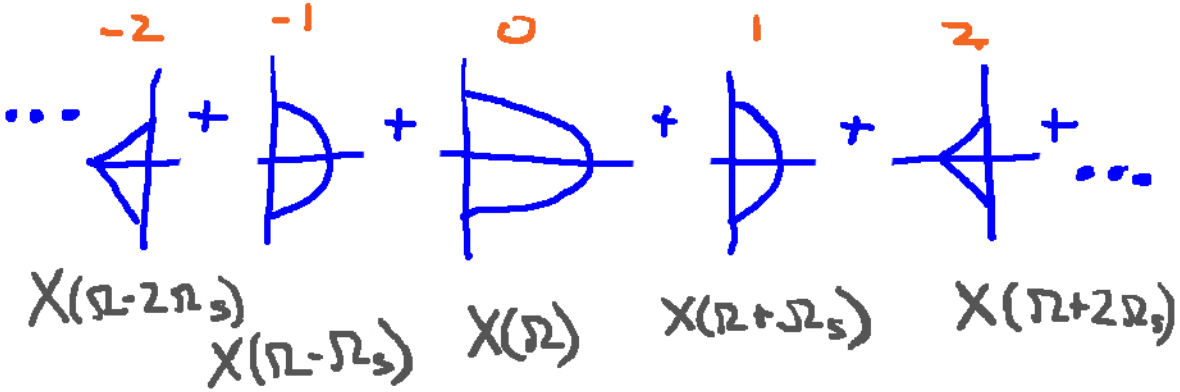


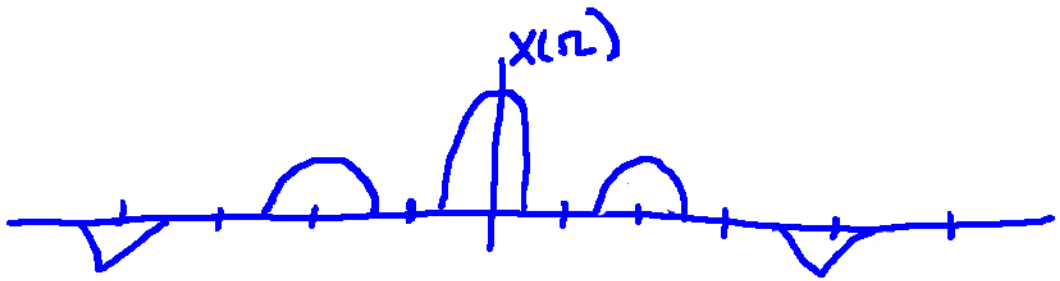
Define $x[n] \overset{FT}{\longleftrightarrow} X_s(\Omega)$

where $X_s(\Omega) = X(e^{j\omega})\Big|_{\omega = \Omega T}$

Since $X(e^{j\omega})$ is 2π periodic, $X_s(\Omega)$ is $\frac{2\pi}{T} = \Omega_s$ periodic





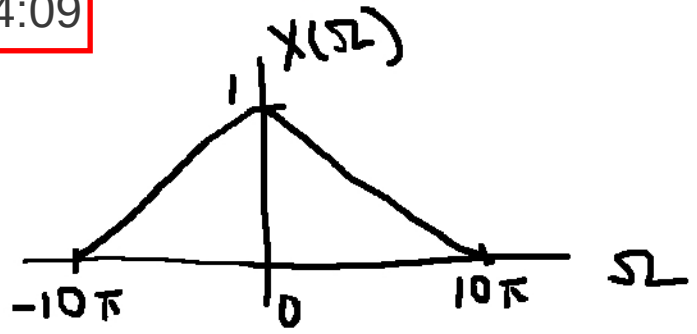


$$X_s(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(\Omega - k\Omega_s)$$

14:09

Example:

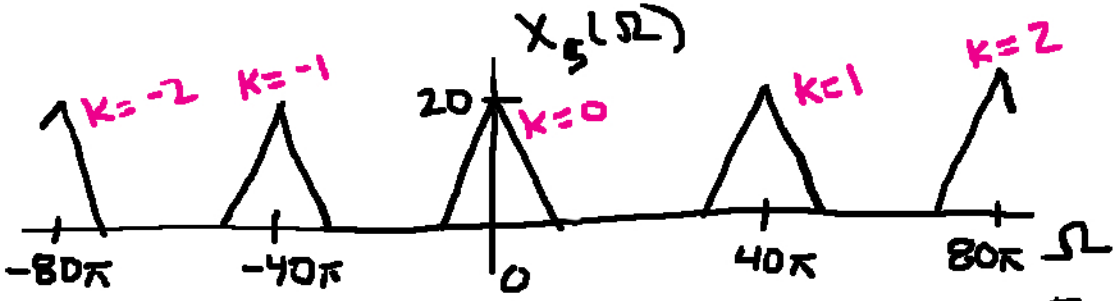
$$X_s(\Omega) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X(\Omega - k\frac{2\pi}{T})$$



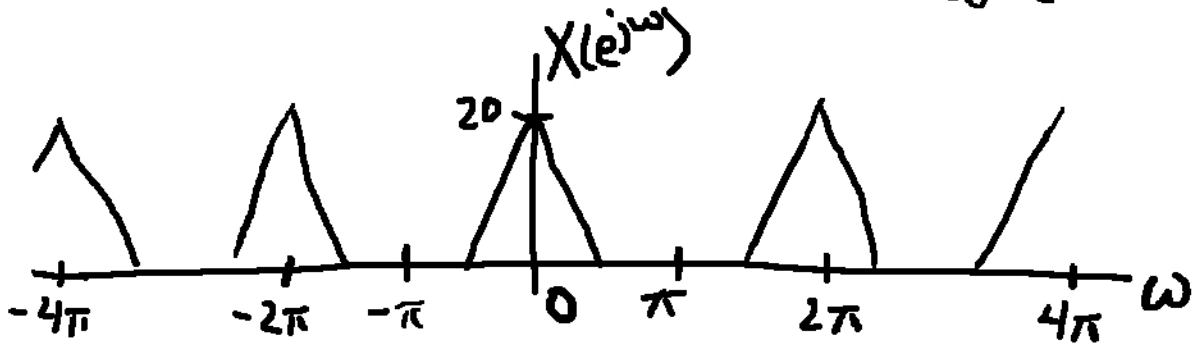
Bandlimited to $W = 10\pi$ rads/sec

Sampling Thm: $\frac{\pi}{T} > 10\pi \Rightarrow T < 0.1$ sec

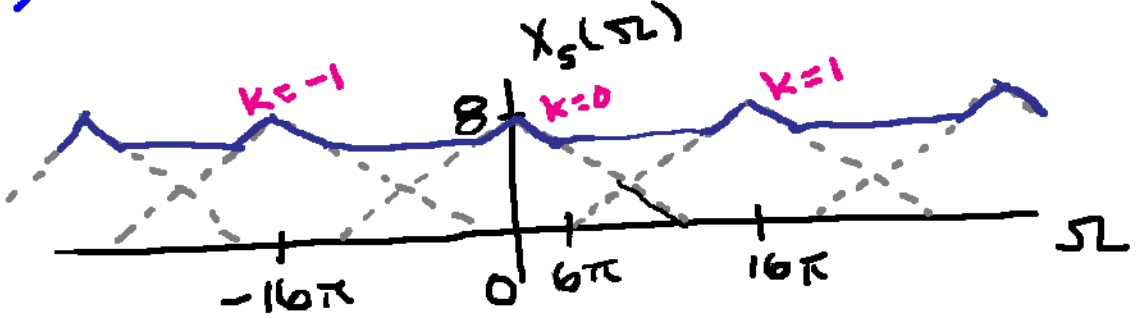
1) $T = 0.05 \Rightarrow \Omega_s = \frac{2\pi}{T} = 40\pi$

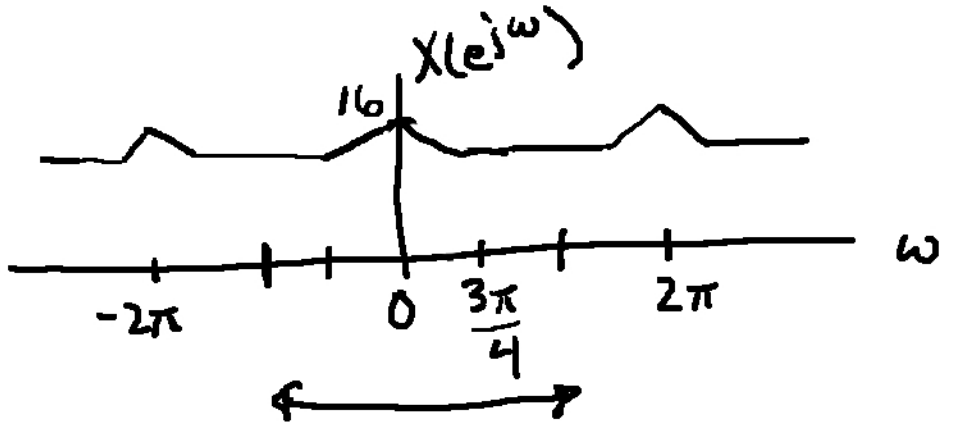


$\omega = \Omega T$



2) $T = 1/8 \Rightarrow \Omega_s = 16\pi$





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